

Billar-Metegol

Simulación dirigida por eventos y búsqueda de la configuración óptima

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Intro.



Impl.



Sim.



Resultados



Conc.

Introducción

Modelo matemático: partículas

- Vuelo libre:

$$x_i(t) = x_i(0) + v_{x_i} t, \quad y_i(t) = y_i(0) + v_{y_i} t$$

- Colisión elástica i - j ($\sigma = R_i + R_j$; $\Delta \mathbf{v} = \mathbf{v}_j - \mathbf{v}_i$, $\Delta \mathbf{r} = \mathbf{r}_j - \mathbf{r}_i$):

$$J = \frac{2 m_i m_j (\Delta \mathbf{v} \cdot \Delta \mathbf{r})}{\sigma (m_i + m_j)}$$

- Actualización post-choque:

$$vx_i^d = vx_i^a + J_x / m_i,$$

$$vy_i^d = vy_i^a + J_y / m_i,$$

$$vx_j^d = vx_j^a - J_x / m_j,$$

$$vy_j^d = vy_j^a - J_y / m_j.$$

Modelo matemático: contorno

- Obstáculo circular fijo ($m \rightarrow \infty$), ($c_n = c_t = 1$):

$$\hat{\mathbf{e}}_n = (\cos \alpha, \sin \alpha), \quad \hat{\mathbf{e}}_t = (-\sin \alpha, \cos \alpha)$$

$$\mathbf{v}^d = \mathbf{R}(-\alpha) \mathbf{S}(c_n, c_t) \mathbf{R}(\alpha) \mathbf{v}^a$$

$$\mathbf{R}(\alpha) = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}, \quad \mathbf{S}(c_n, c_t) = \begin{pmatrix} -c_n & 0 \\ 0 & c_t \end{pmatrix}$$



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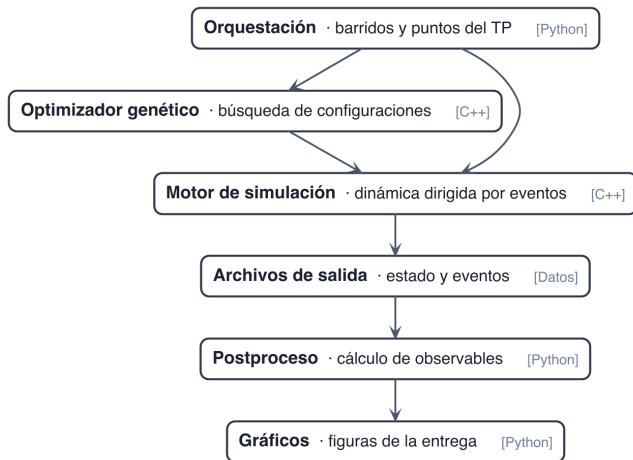
Resultados



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Implementación

Arquitectura del código





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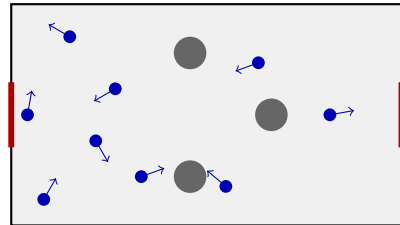


Conc.

Simulaciones

Geometría del sistema

- ▶ Mesa: $L \times W = 1.20 \times 0.68 \text{ m}^2$
- ▶ Arco: largo $d = 0.20 \text{ m}$
- ▶ N partículas: $r = 0.0175 \text{ m}$, $m = 0.025 \text{ kg}$
- ▶ $v_0 = 1 \text{ m/s}$
- ▶ K obstáculos circulares fijos con $K \geq 1$



Observables

- ▶ $N_g(t)$: partículas que impactaron con un arco.
- ▶ $F_u(t) = N_g(t)/N$
- ▶ t_{90} : primer instante con $F_u(t) \geq 0.9$.
- ▶ $\langle t_{90} \rangle_n$: promedio de t_{90} sobre n realizaciones.

$$\text{DCM}(t) = \frac{1}{N} \sum_{i=1}^N |\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2 \approx 4 D t,$$

con D el coeficiente de difusión.

Configuraciones estudiadas

Configuración	Obstáculos	Método	Se barre
Mesa vacía	ninguno	baseline	—
Círculo	1, centrado	barrido exhaustivo	radio R
Quad	K c. simetría quad	Algoritmo genético (GA)	$K \in \{5, 9, 13, 17\}$
Pared	pared de círculos	Algoritmo genético (GA)	grano, puntos de perfil

Algoritmo genético

Fitness
$$f(s) = \begin{cases} t_{90}(s) & F_u \geq 0.9 \\ t_{\text{máx}} + \alpha (90 - N_g(t_{\text{máx}})) & \text{si no} \end{cases}$$

$$\langle t_{90} \rangle_5 = \frac{1}{5} \sum_{j=1}^5 f(s_j),$$

Selección Torneo de 3

Cruza BLX- α : $g_h \sim U[g_{\text{mín}} - 0.3 \Delta, g_{\text{máx}} + 0.3 \Delta]$, $\Delta = g_{\text{máx}} - g_{\text{mín}}$

Mutación $g' = g + \mathcal{N}(0, \sigma^2)$ con prob. 0.3 por gen, σ decrece con las generaciones

Elitismo 2 mejores



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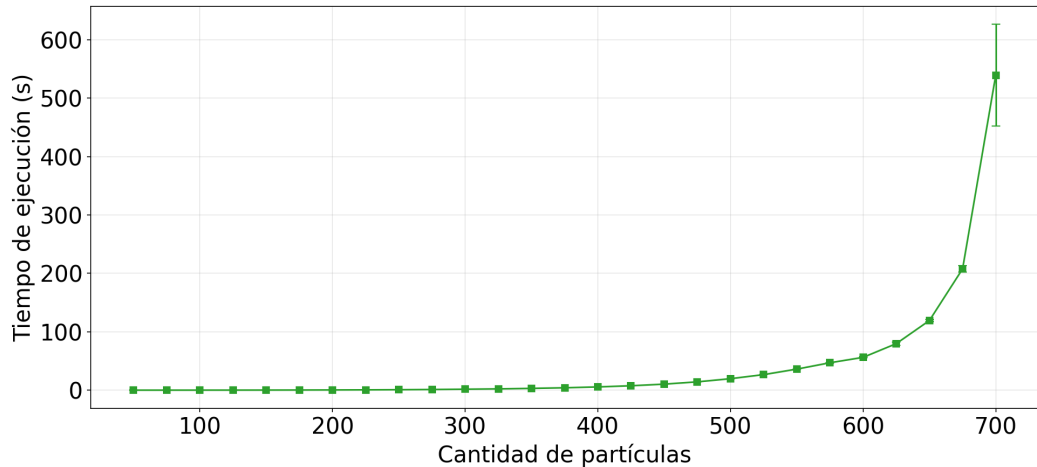


Conc.

Resultados

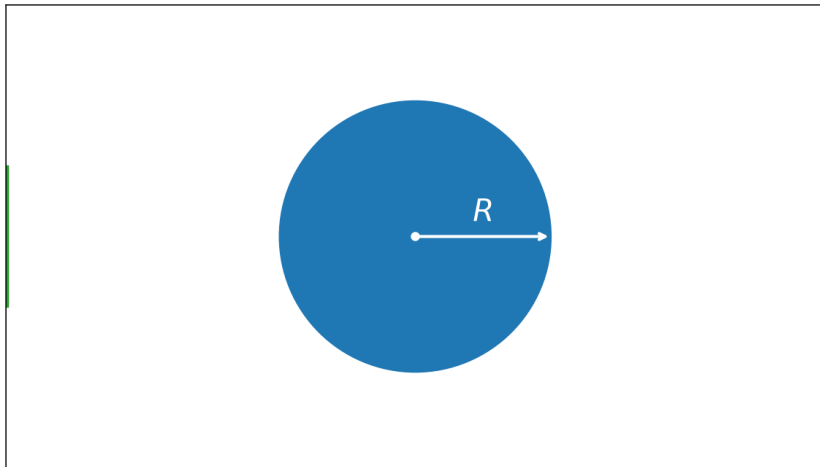
Tiempo de ejecución vs Cantidad de partículas

Sin obstáculos, $t_f = 30$ s

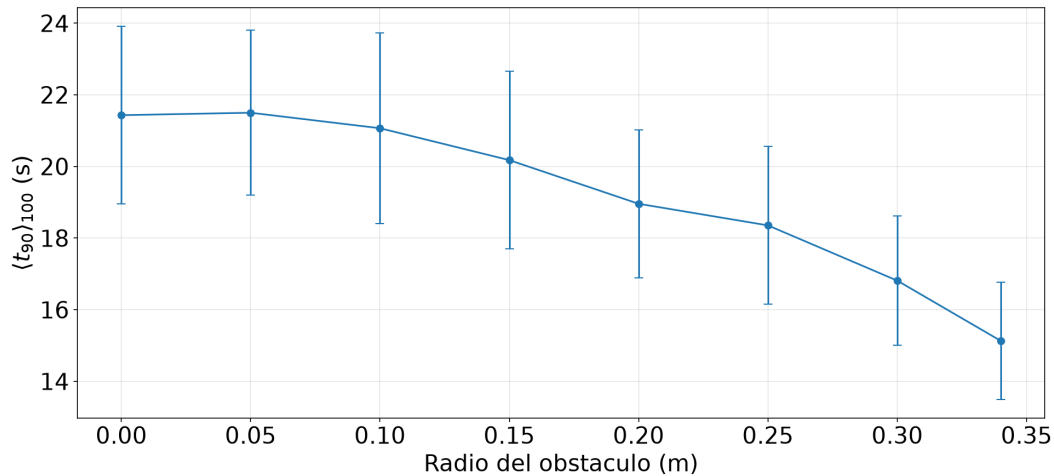


Círculo: codificación

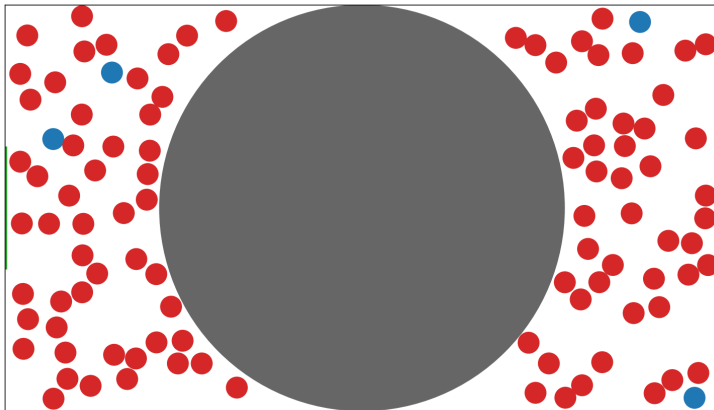
$$R \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.34\} \text{ m}$$



$\langle t_{90} \rangle_{100}$ vs Radio del círculo



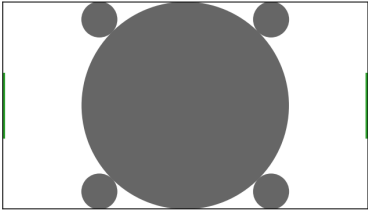
Círculo: mejor resultado



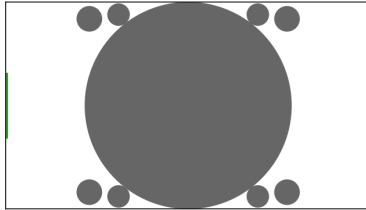
Video: <https://youtu.be/TODO>

Quad: geometrías propuestas por el GA

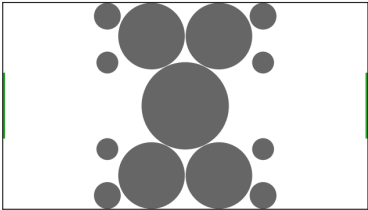
K=5 $\langle t_{90} \rangle_{100} = 15.70 \pm 2.11$ s



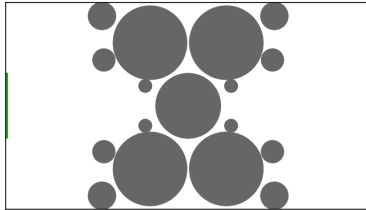
K=9 $\langle t_{90} \rangle_{100} = 14.25 \pm 1.69$ s



K=13 $\langle t_{90} \rangle_{100} = 16.65 \pm 2.08$ s



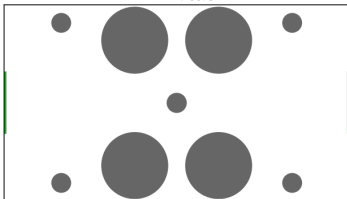
K=17 $\langle t_{90} \rangle_{100} = 16.40 \pm 2.00$ s



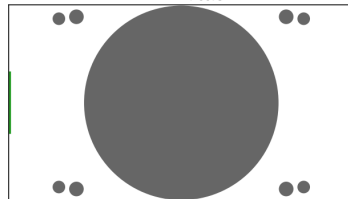
Quad: evolución del ganador

 $K = 9$

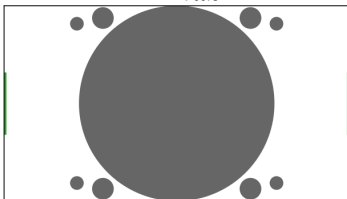
Generacion 0 $\langle t_{90} \rangle_5 = 18.74$ s



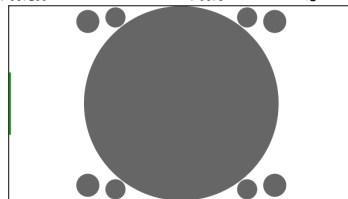
Generacion 20 $\langle t_{90} \rangle_5 = 14.15$ s



Generacion 40 $\langle t_{90} \rangle_5 = 14.08$ s

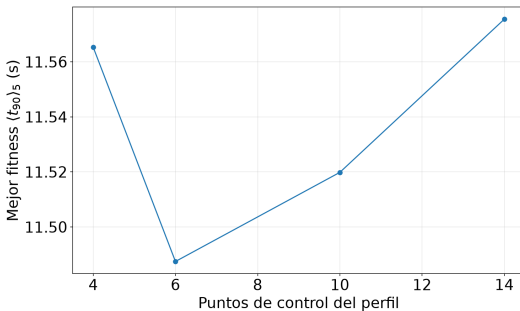
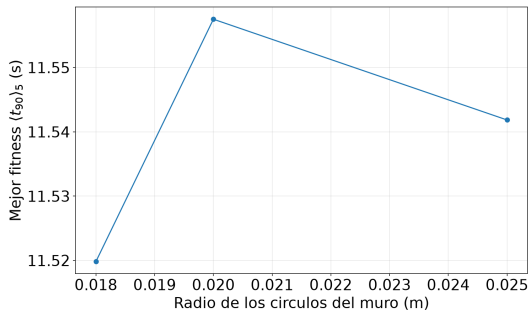


Ganador $K=9$
 $\langle t_{90} \rangle_{100} = 14.25 \pm 1.69$ s $\langle t_{90} \rangle_5 = 12.79$ s (gen 58)



Pared: hiperparámetros

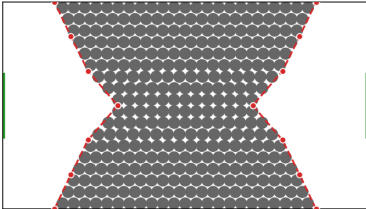
Pobl. 40, 80 gen.



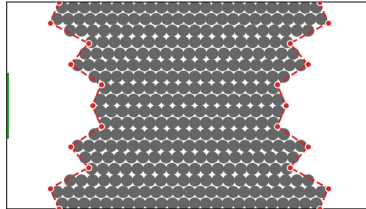
Pared: geometrías por puntos de perfil

Grano 0.018 m, pobl. 40, 80 gen.

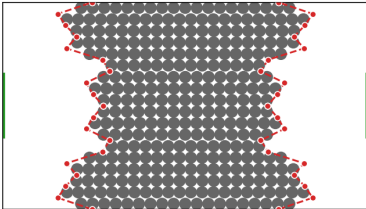
perfil=4 $\langle t_{90} \rangle_{100} = 13.83 \pm 1.69$ s



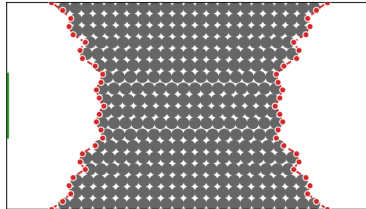
perfil=6 $\langle t_{90} \rangle_{100} = 14.42 \pm 1.78$ s

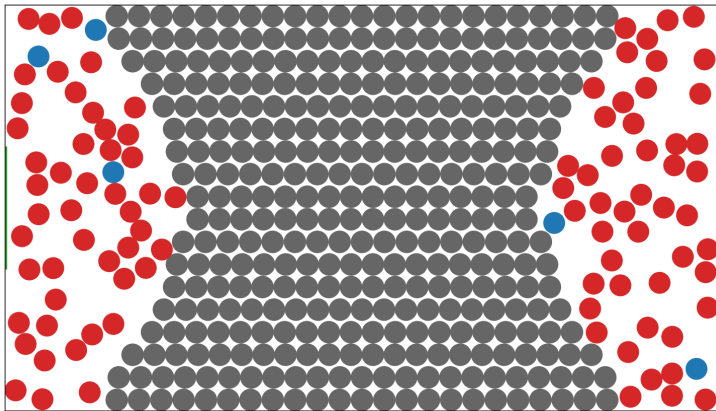


perfil=10 $\langle t_{90} \rangle_{100} = 13.89 \pm 1.55$ s



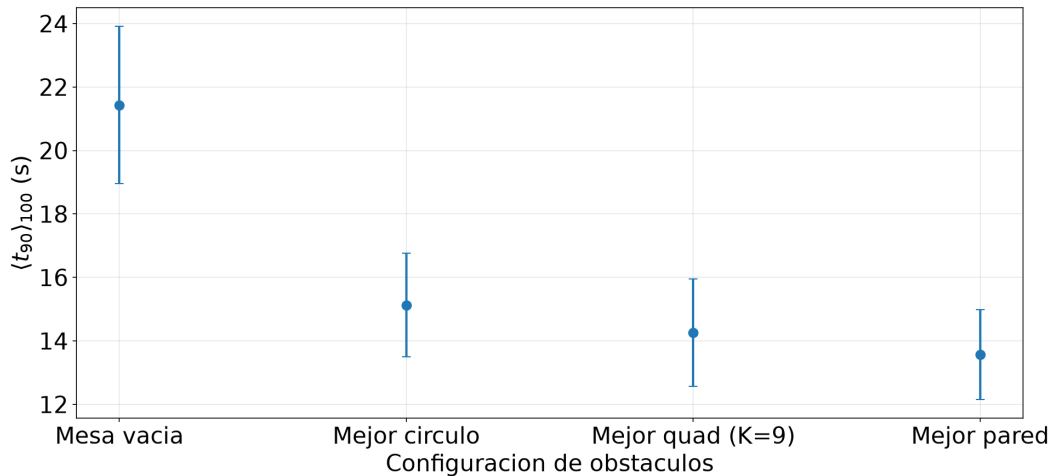
perfil=14 $\langle t_{90} \rangle_{100} = 13.89 \pm 1.63$ s



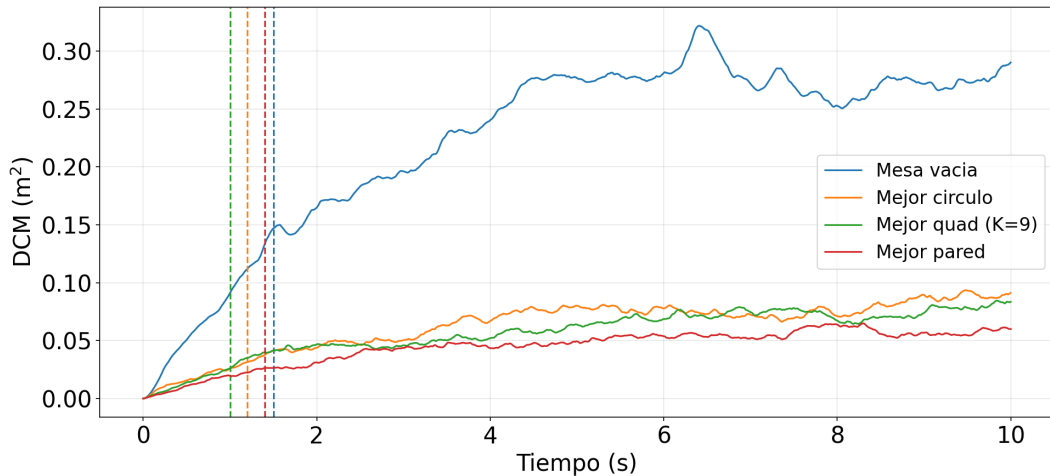


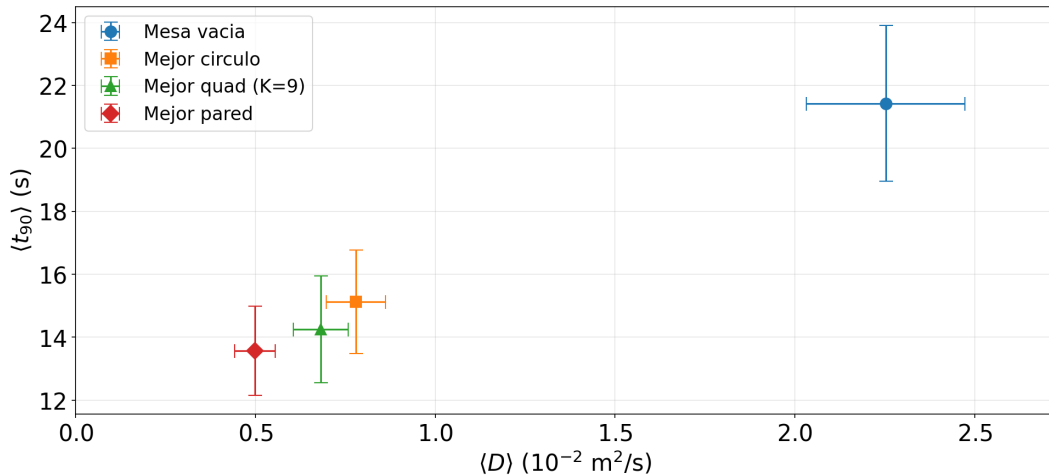
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Comparación final



DCM vs Tiempo



$\langle D \rangle$ vs $\langle t_{90} \rangle$ 



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Configuración elegida

Competencia



Conclusiones



Conclusiones





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Gracias!